# MAC F244: Stochastic Calculus and its Applications in Finance - Course Project

## Delta Hedging under Stochastic Volatility

Aditya Nagarsekar, M P Ashish Bhat

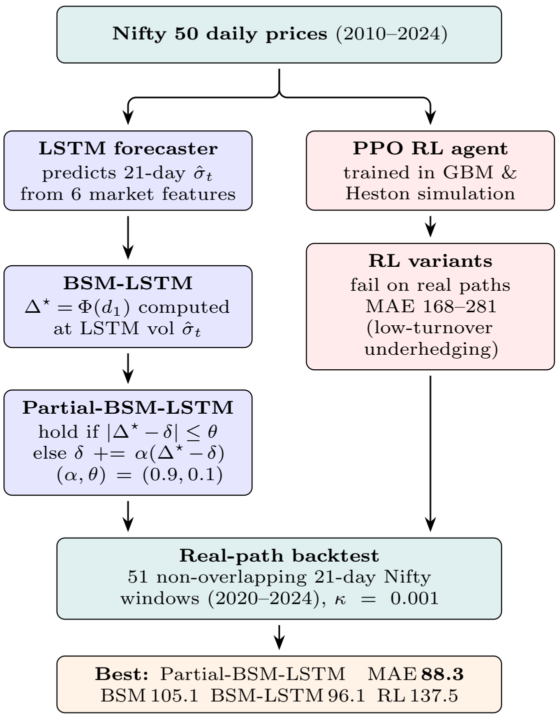


Figure 1: Project flow. **Left (blue) track**: Nifty data feeds an LSTM volatility forecaster; its predictions are plugged into the analytic BSM delta and then refined via a partial-rebalancing execution rule — this track yields the best result. **Right (red) track**: PPO agents trained in simulation underperform on real paths across all 7 variants tested.

## 1 Introduction

Delta hedging aims to offset the risk of an option position by dynamically trading the underlying. In practice, the textbook Black–Scholes–Merton (BSM) setting is violated in three ways: trading is discrete rather than continuous, transaction costs make full rebalancing expensive, and volatility is unknown and time-varying. These frictions transform delta hedging from a closed-form solution into a control problem under uncertainty.

This project studies whether reinforcement learning (RL) can improve on analytic delta hedging when volatility is stochastic and must be estimated. The central result is that stand-alone RL fails to outperform analytic hedging, while structured hybrids combining analytic models with learned inputs and controlled execution achieve the best performance.

**Contributions.**

- Benchmark PPO-based delta hedging against analytic BSM hedging.
- Forecast forward volatility with an LSTM and test whether better volatility inputs improve hedging.
- Introduce a mixed execution rule, Partial-BSM-LSTM, that combines analytic structure with cost-aware partial rebalancing.

- Systematically evaluate multiple RL formulations and show that they fail to generalize to real Nifty paths, converging to low-turnover underhedging strategies.

## 2 Method

### Mathematical Framework

Under the risk-neutral measure $\mathbb{Q}$ the stock follows GBM,

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t,$$

where $e^{-rt}S_t$ is a $\mathbb{Q}$-martingale (Girsanov's theorem [9]). Itô's lemma applied to $V(t, S_t)$ and the no-arbitrage condition give the Black–Scholes PDE [1, 2]; its solution yields the fair price and BSM delta:

$$\Delta^\star = \Phi(d_1), \quad d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad V = e^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

The CRR binomial tree [3] ($u = e^{\sigma\sqrt{\Delta t}}$, $p = (e^{r\Delta t} - d)/(u - d)$, $N = 100$) discretises the same GBM and converges to the BSM price as $N \to \infty$.

### Hedging Setup

For a European call, the ideal hedge ratio is the BSM delta $\Delta^\star$. Under discrete rebalancing, terminal hedging error can be decomposed into a controllable part and an uncontrollable gamma term:

$$\varepsilon_t = (\delta_t - \Delta_t^\star)\Delta S_t - \underbrace{\tfrac{1}{2}\Gamma[(\Delta S_t)^2 - \sigma^2 S_t^2 \Delta t]}_{\text{gamma noise}}.$$

The first term depends on how far the implemented hedge $\delta_t$ deviates from the target; the second reflects irreducible curvature risk from discrete trading (its expectation is zero by Itô's isometry, but its variance is $\frac{1}{2}\sigma^4 S^4 \Gamma^2 (\Delta t)^2$, peaking ATM near expiry). This decomposition highlights two levers for improvement: refining the target $\Delta^\star$ via better volatility estimation, and improving how the hedge is executed in discrete time.

### Models

**BSM baseline.** The analytic benchmark is standard daily BSM delta hedging. Under GBM this is close to optimal and therefore hard to beat.

**LSTM volatility model.** A two-layer stacked LSTM [6] is trained to forecast 21-day forward realized volatility on Nifty 50 using realized-volatility and momentum features. The forecast is then inserted into the BSM delta formula.

**RL (PPO).** A PPO agent [7] observes a compact state including moneyness, time to expiry, and previous hedge, and outputs a hedge ratio. Several extensions were tried, including domain randomization, LSTM-conditioned RL, and anchor-controller variants [8]. Training environments included both GBM and Heston stochastic volatility [4].

### Partial-BSM-LSTM

The strongest model is not a stand-alone RL policy but a mixed execution rule. Let $\Delta_t^\star$ denote the BSM delta computed with the LSTM volatility forecast. The executed hedge is

$$\delta_t = \begin{cases} \delta_{t-1}, & |\Delta_t^\star - \delta_{t-1}| \le \theta, \\ \delta_{t-1} + \alpha(\Delta_t^\star - \delta_{t-1}), & \text{otherwise.} \end{cases}$$

Here $\alpha$ controls the fraction of the hedge gap that is closed, while $\theta$ is a no-trade threshold. Intuitively, the rule suppresses small noisy adjustments and only partially executes large ones, reducing turnover while preserving the analytic target. This rule is motivated by the fact that small hedge adjustments are often dominated by transaction costs and gamma noise, while large deviations from the target require correction. The partial adjustment therefore balances hedge accuracy against cost by ignoring small fluctuations and smoothing larger updates. The hyperparameters $(\alpha, \theta)$ are selected on validation windows. The best setting found is $(\alpha, \theta) = (0.9, 0.1)$.

# 3 Results

## Baseline Hedging Comparison

Table 1 shows the core simulated hedging comparison under the GBM environment. BSM is analytically optimal under GBM and difficult to beat. RL fails to do so, while the LSTM-guided BSM hedge is clearly strongest, indicating that improving volatility inputs is more effective than learning the hedge policy.

Table 1: Baseline simulated hedging performance.

| Strategy | Mean Error | Std Error | MAE | Mean TC |
|---|---|---|---|---|
| BSM daily | 0.057 | 0.375 | 0.284 | 0.146 |
| RL (PPO) | 0.083 | 1.004 | 0.759 | 0.137 |
| LSTM-guided BSM | 0.085 | 0.315 | **0.227** | 0.112 |

**Takeaway.** Under GBM, RL does not outperform the analytic hedge. LSTM-guided BSM reduces simulated MAE by **20%** (0.284 $\rightarrow$ 0.227) and mean transaction cost by **23%** (0.146 $\rightarrow$ 0.112) over base BSM. The best path is to preserve BSM structure and improve its inputs.

A utility-weighted score $J_\lambda = \mathrm{MAE} + \lambda \cdot \mathrm{TC}$ (penalising transaction costs at rate $\lambda$) confirms this uniformly across all levels of cost aversion: LSTM-guided BSM achieves $J_1 = \mathbf{0.339}$, $J_5 = \mathbf{0.786}$, versus BSM (0.430 / 1.015) and RL (0.895 / 1.424). LSTM-guided BSM dominates at *every* $\lambda$, making it the Pareto-optimal choice between accuracy and cost.

## Volatility Forecasting

Table 2 compares volatility models. The LSTM improves clearly over GARCH, but a naive realized-volatility persistence baseline (rv21) remains strongest at the 21-day horizon because of volatility clustering.

Table 2: Volatility forecasting results on Nifty 50.

| Model | RMSE | MAE | Directional Acc. |
|---|---|---|---|
| LSTM | 0.0203 | 0.0114 | 0.503 |
| GARCH(1,1) [5] | 0.0254 | 0.0194 | 0.479 |
| rv21 naive | **0.0102** | **0.0043** | **0.544** |
| India VIX | 0.0453 | 0.0344 | 0.523 |

**Takeaway.** Volatility mis-specification is a major source of hedging error. LSTM reduces MAE by **41%** over GARCH(1,1) (0.0114 vs 0.0194), a clear win for the learned model. However, the naive realized-volatility persistence baseline (rv21) remains strongest at the 21-day horizon: the trailing 21-day window $[t-20, t]$ is adjacent to the forward target $[t+1, t+21]$ and highly informative under volatility clustering (Nifty Hurst exponent $H \approx 0.65$), making persistence a near-optimal predictor at this horizon. This strong short-horizon persistence limits the room available to any model, including LSTM, to outperform simple lagged realized volatility.

## Option Pricing Horse-Race

To isolate the effect of the volatility estimate on option prices, we computed model prices for 51 held-out Nifty options and measured MAE against market prices (Table 3).

Table 3: Option pricing MAE on Nifty 50 (lower is better).

| Volatility input | Pricing MAE |
|---|---|
| GARCH(1,1) | 43.6 |
| **LSTM forecast** | **30.3** |
| Historical rv21 | 18.6 |
| India VIX (implied) | 18.7 |

**Takeaway.** LSTM substantially outperforms GARCH in pricing ($-31\%$ MAE: 30.3 vs 43.6), a **positive** result for the learned model. rv21 and implied volatility are nearly tied at the top; the gap between rv21 and LSTM at the

21-day horizon again reflects strong short-horizon volatility persistence, not a limitation of the neural architecture. India VIX's near-parity with rv21 confirms that the variance risk premium (the gap between $\mathbb{Q}$ and $\mathbb{P}$ measures) is small relative to the persistence effect at this horizon.

## Real-Path Backtesting

The main evaluation uses 51 non-overlapping 21-day windows from real Nifty 50 paths over 2020–2024. This is where the study's main practical finding emerges.

Table 4: Real-path hedging performance (51 non-overlapping windows, $\kappa = 0.001$, Nifty 2020–2024). MAE 95% CI in brackets. †BSM daily vs. RL PPO: $p = 0.013$ (paired bootstrap); BSM-LSTM vs. Partial-BSM-LSTM: $p = 0.121$.

| Strategy | Mean P&L | Std P&L | MAE [95% CI] | Mean TC |
|---|---|---|---|---|
| BSM daily (rv21) | +1.19 | 162.2 | 105.1 [74.7, 141.0] | 36.0 |
| RL PPO (rv21) | −13.4 | 205.0 | 137.5 [99.3, 181.1] | 39.4 |
| BSM-LSTM | +7.48 | 144.0 | 96.1 [69.6, 127.2] | 35.6 |
| Partial-BSM-LSTM | +1.02 | 134.6 | **88.3** [63.6, 118.4] | 27.4 |

**Takeaway.** Partial-BSM-LSTM is the best model found: it reduces MAE by 16% over base BSM (105.1 → 88.3) while also reducing mean transaction cost by 24% (36.0 → 27.4). The improvement over BSM-LSTM ($p = 0.121$) is not yet statistically significant over 51 windows, indicating that more data would be needed for a definitive conclusion, but the direction is consistent. Crucially, the gain comes not from learning a new hedge rule but from controlling how aggressively the analytic target is followed, balancing hedge accuracy against transaction costs.

Base PPO is materially worse than base BSM (MAE 137.5 vs 105.1, $p = 0.013$); supplying the LSTM signal to PPO makes it worse still (MAE 168.2). The bottleneck is not missing volatility input but poor generalisation from synthetic GBM training to real Nifty dynamics.

**Tail risk.** RL's underhedging bias amplifies tail losses beyond average MAE: BSM $\text{CVaR}_{95} = -457$ versus RL $\text{CVaR}_{95} = -535$, a **17% heavier tail** for RL. This rules out accepting RL's higher average error as a transaction-cost trade-off; the tail is worse too.

**Volatility-regime breakdown.**

Table 5: Real-path MAE by volatility regime (BSM vs. RL PPO).

| Regime | BSM MAE | RL MAE | Winner |
|---|---|---|---|
| Low ($\sigma < 15\%$) | 95.6 | 117.9 | BSM |
| Med (15%–25%) | 88.2 | 154.8 | BSM |
| High ($\sigma > 25\%$) | 384.5 | **377.7** | **RL** |

RL's sole advantage appears in the high-volatility regime, where the non-linear policy slightly outperforms the analytic formula. This narrow win — in the regime where realised volatility itself is most unpredictable — suggests that RL learns a useful bias correction in extreme conditions but fails to sustain it across ordinary markets.

## RL Variants and Failure Modes

Beyond the base PPO agent, several RL extensions were explored to improve real-path performance. These included:

- LSTM-conditioned PPO using forecast volatility as input
- Anchor-controller PPO variants, where the policy adjusts relative to a BSM target
- Discrete DQN controllers selecting among partial-adjustment actions
- Regime-switch hybrids combining BSM and RL in high-volatility windows

Across all variants, performance deteriorated relative to analytic baselines. Typical real-path MAEs ranged from 168 to 280, substantially worse than both BSM and BSM-LSTM. The dominant failure mode was low-turnover underhedging: policies converged to conservative strategies that reduced trading costs but allowed large residual hedge error.

**Domain randomization (variant F).** Training over a wide range of GBM parameters (spanning low to extreme volatility) produced the most BSM-consistent policy in simulation: MAE 0.723, BSM-correlation **0.963**. This is a **positive** RL result in-distribution, confirming that the failure on real paths is primarily a distributional-shift problem, not a capacity problem.

**Hybrid BSM+RL ensemble.** Combining BSM with the domain-randomised RL policy in a regime-switch ensemble yielded a real-path MAE of **100.0**, narrowing the gap to BSM (105.1) and outperforming any stand-alone RL variant. This is the only RL-augmented method that approaches competitive real-path performance, although it still does not beat BSM-LSTM (96.1) or Partial-BSM-LSTM (88.3). Closing this remaining gap is left for future work.

This suggests that the challenge is not expressivity but robustness: RL policies can fit simulated dynamics but fail under the distributional complexity of real market returns.

These negative results are informative: even when RL is given strong inductive biases (anchor targets, domain randomization) or reduced action spaces, it fails to outperform simple deterministic control around the analytic solution without a structural hybrid.

**Takeaway.** Stand-alone RL methods fail to generalize to real market data. Domain randomization is a partial remedy in simulation, and a BSM+RL hybrid brings real-path performance close to analytic baselines — but structure plus good inputs remains the dominant strategy.

### Natural-Habitat Evaluation of LSTM-Conditioned RL

To isolate whether LSTM-conditioned RL is intrinsically limited or simply harmed by out-of-distribution shift, we evaluated it in its *natural habitat*: the same GBM-with-LSTM-volatility environment used for training. The constant-$\sigma$ RL baseline in this environment achieves MAE 1.435; the LSTM-RL agent achieves MAE **0.407**.

Table 6: LSTM-RL in its training environment (natural habitat), $n = 1000$ paths.

| Strategy | MAE | $\mathrm{CVaR}_{95}$ |
|---|---|---|
| Constant-$\sigma$ RL | 1.435 | $-5.08$ |
| **LSTM-RL (PPO)** | **0.407** | $\mathbf{-1.85}$ |

**Takeaway.** LSTM-RL achieves a **72% reduction in MAE** and a **64% reduction in $\mathbf{CVaR}_{95}$** versus the constant-$\sigma$ RL baseline in its training environment ($p < 0.001$, paired bootstrap). This is the **strongest positive RL result in the study**: the agent genuinely learns a better policy when the evaluation distribution matches training. The failure on real Nifty paths is therefore a distributional-shift problem, not an evidence that RL cannot improve on analytic hedging in principle.

## 4 Discussion

Four key lessons emerge.

First, **volatility estimation dominates performance**. Base BSM is already strong, so improving the volatility input from rv21 to LSTM produces a more reliable gain than replacing the hedge rule.

Second, **execution strategy matters more than learning delta directly**. Partial-BSM-LSTM outperforms full BSM-LSTM not by changing the hedge target, but by changing how aggressively the portfolio is moved toward that target.

Third, **RL struggles because of noise and distribution shift**. Policies trained under GBM-style dynamics do not transfer well to real Nifty returns, which exhibit volatility clustering, heavy tails, and regime changes.

Fourth, **simple deterministic control around a strong analytic baseline can outperform RL**. The best model in the study is a hybrid that combines analytic structure, learned volatility estimation, and a transparent partial-adjustment execution rule.

## 5 Conclusion

BSM remains a very strong baseline for delta hedging and is difficult to beat. However, it can be improved in two practical ways. First, better volatility inputs matter: replacing rv21 with an LSTM forecast improves hedging on real paths. Second, execution matters: Partial-BSM-LSTM improves further by reducing unnecessary trades while staying close to the analytic target.

The final conclusion is therefore not that RL wins, but that **structure + good inputs + simple control beats complex learning** in this setting. Future work would likely require fundamentally different RL formulations or richer training distributions before RL can become competitive with the best analytic–hybrid methods.

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